Empirical Properties of Directed Acyclic Graphs in Natural Language

Project LE2: Linguistic Empiricism

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1. Motivation

The syntactic structure of a sentence can be represented as a dependency tree, a directed acyclic graph (DAG), where words are nodes and grammatical relations are directed edges from head to dependent. Such representations are central to computational linguistics, psycholinguistics, and theories of language processing. A fundamental question in this space is whether the structural properties observed in natural language dependency trees are arbitrary or reflect systematic constraints shaped by cognition and grammar. Random trees offer a natural baseline: given a sentence of n words, what would its dependency structure look like if edges were assigned at random rather than by linguistic rules? If human language trees look like random trees, their structure carries no special information. If they differ systematically, that divergence reveals the fingerprint of linguistic constraints. This project empirically investigates these structural differences across ten typologically diverse languages.

Research Question: What structural properties characterise dependency graphs in human language, and how do these properties differ from randomly generated directed acyclic graphs with the same number of nodes?

2. Hypotheses and Predictions

We test three hypotheses, motivated by cognitive constraints on language production and comprehension:

H1 (Arity): The distribution of node out-degrees in real trees will differ from random trees even though mean arity is identical, because linguistic rules create polarised structures. Many leaf nodes and a few high-arity heads rather than uniform branching.

H2 (Depth): Real dependency trees will be significantly shallower than random trees of the same size, reflecting working memory constraints that discourage deeply nested syntactic structures.

H3 (Density): Since both real and random structures are valid trees with n nodes and $n-1$ edges, graph density is identical across conditions by mathematical necessity.

3. Methods

3.1 Data. Dependency-parsed sentences were obtained from the Surface-Syntactic Universal Dependencies (SUD) corpus v2.17, downloaded from grew.fr. We used the training split of treebanks from eleven typologically diverse languages: English, Hindi, French, German, Spanish, Chinese, Arabic, Russian, Japanese, Turkish, and Basque. Sentences with multiword tokens, empty nodes, or missing roots were excluded. After filtering, each language contributed between 1,200 and 15,000 valid sentence trees.

3.2 Graph Representation. Each sentence was converted to a directed graph $G = (V, E)$ using NetworkX, where V is the set of word tokens and each edge (h, d) indicates that word h is the syntactic head of dependent d. Every valid dependency tree has exactly one root node (in-degree 0) and is weakly connected.

3.3 Random Tree Generation. For each real sentence tree of size n, a matched random tree was generated using **Prüfer sequences**. A Prüfer sequence of length $n-2$ drawn uniformly from $\{0, \ldots, n-1\}$ encodes a uniformly random labeled tree on n nodes via the bijection:

$$\text{Prüfer}(T) : \text{labeled trees on } n \text{ nodes} \longleftrightarrow \{0, \ldots, n-1\}^{n-2}$$

The resulting undirected tree is rooted at node 0 and oriented via BFS to produce a directed tree, ensuring every random tree has the same node count as its paired real sentence.

3.4 Structural Metrics. Three metrics were computed for each tree:

1. **Arity**: Out-degree of each node (number of direct dependents). Mean arity $= (n-1)/n$ for any tree, so distributional *shape* is what matters.
2. **Tree Depth**: The length of the longest path from the root to any leaf, computed via BFS. This measures how deeply nested the structure is.
3. **Graph Density**: Define $\rho = |E|/(|V| \cdot (|V|-1))$. For a tree this equals $1/n$, so density is entirely determined by sentence length and is identical for real and random trees of equal size.

3.5 Statistical Inference. The two-sample Kolmogorov-Smirnov (KS) test was used to compare real and random distributions. The KS statistic $D = \sup_x |F_1(x) - F_2(x)|$ measures the maximum absolute difference between the two empirical CDFs. Results with $p < 0.05$ are deemed statistically significant.

3.6 Worked Example. Consider the sentence *"The cat sat on the mat"* ($n = 6$ words). The SUD annotation produces the following dependency tree, rooted at *sat*:

ID	Form	Head	DepRel	Out-degree
1	The	2	det	0
2	cat	3	nsubj	1
3	sat	0	root	2
4	on	3	obl	1
5	the	6	det	0
6	mat	4	nmod	1

The three metrics are computed as follows:

Arity. The out-degree sequence is $\{0, 1, 2, 1, 0, 1\}$. Mean arity $= 5/6 = (n-1)/n$, confirming the identity holds. The distribution is already polarised: two leaf nodes (arity 0), three nodes with exactly one child, and the root *sat* with arity 2.

Depth. Starting from the root *sat* (depth 0), the longest path follows *sat* $\rightarrow$ *on* $\rightarrow$ *mat*, giving depth $= 2$.

Density.

$$\rho = \frac{|E|}{|V| \cdot (|V| - 1)} = \frac{5}{6 \times 5} = \frac{1}{6} \approx 0.167$$

which equals $1/n = 1/6$, confirming that density is entirely determined by sentence length.

A matched random tree on 6 nodes generated via Prüfer sequence might produce the structure *sat* $\rightarrow$ *The* $\rightarrow$ *on* $\rightarrow$ *mat* $\rightarrow$ *cat* $\rightarrow$ *the*, yielding depth $= 5$, more than twice the real tree's depth of 2 from the same 6 nodes. This illustrates the core finding: real dependency trees are structurally constrained to be far shallower than random trees of equal size.

4. Results

4.1 Arity Distribution. Across all languages, mean arity is identical between real and random trees (both ≈ 0.91–0.97), as expected. However, KS tests reveal highly significant distributional differences in all languages ($p \approx 0$). As shown in Figure 1, real trees (blue) concentrate more mass at arity 0, more pure leaf nodes and show a steeper dropoff at higher arity compared to random trees (red). This reflects the categorical nature of grammar: most words are syntactic leaves, while a small class of predicates governs many dependents.

4.2 Tree Depth. The depth comparison yields the strongest and most consistent finding. Real trees are significantly shallower than their size-matched random counterparts in every language (all $p \approx 0$, KS stat 0.112–0.406). Depth reductions range from 3% (Japanese) to 33% (Chinese), as shown in Table 1 and Figures 2–3. Japanese is a notable outlier: its head-final, right-branching typology naturally produces deeper dependency chains, narrowing the gap to random trees.

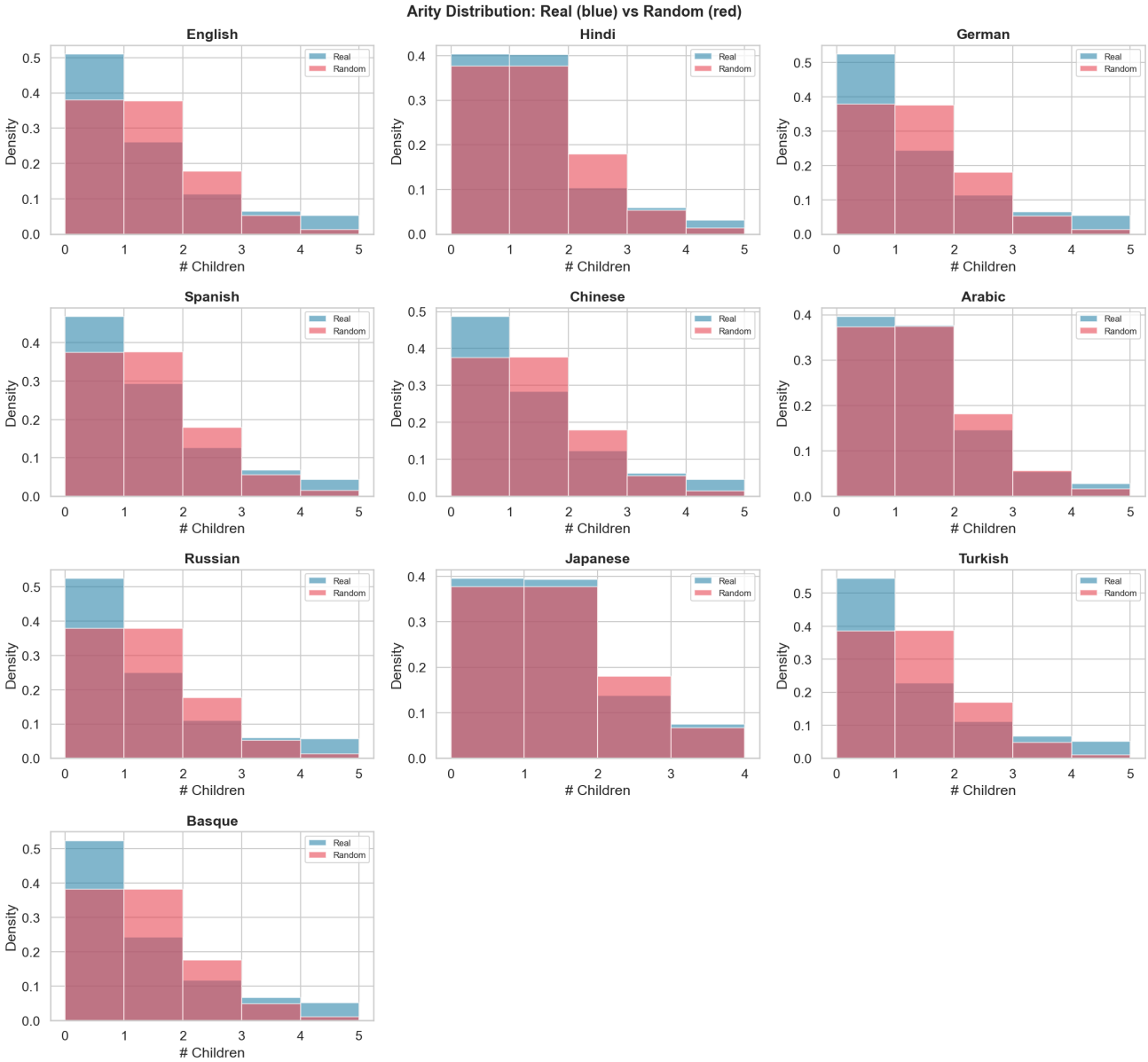

Figure 1: Arity distributions across 10 languages. Real trees (blue) show greater concentration at 0 children and steeper decay than random trees (red).

Table 1: Mean tree depth for real vs. random trees. All results significant at $p < 0.001$ except Japanese ($p = 0.005$).

Language	Real Mean	Random Mean	Reduction	KS Stat
English	5.52	7.09	22%	0.184
Hindi	7.31	8.27	12%	0.118
German	5.38	7.62	29%	0.320
Spanish	7.54	9.51	21%	0.222
Chinese	6.14	9.13	33%	0.406
Arabic	9.98	11.35	12%	0.112
Russian	5.07	7.06	28%	0.247
Japanese	8.50	8.72	3%	0.028
Turkish	3.51	5.07	31%	0.242
Basque	4.21	6.05	30%	0.330

4.3 Graph Density. As predicted by H3, graph density was identical between real and random trees in every language (KS stat $= 0$, $p = 1.0$). Since both are valid trees with n nodes and $n-1$ edges, $\rho = 1/n$ in both cases. Density is not an informative discriminator here; it would only distinguish structures if the comparison involved non-tree graphs that permit multiple heads per node.

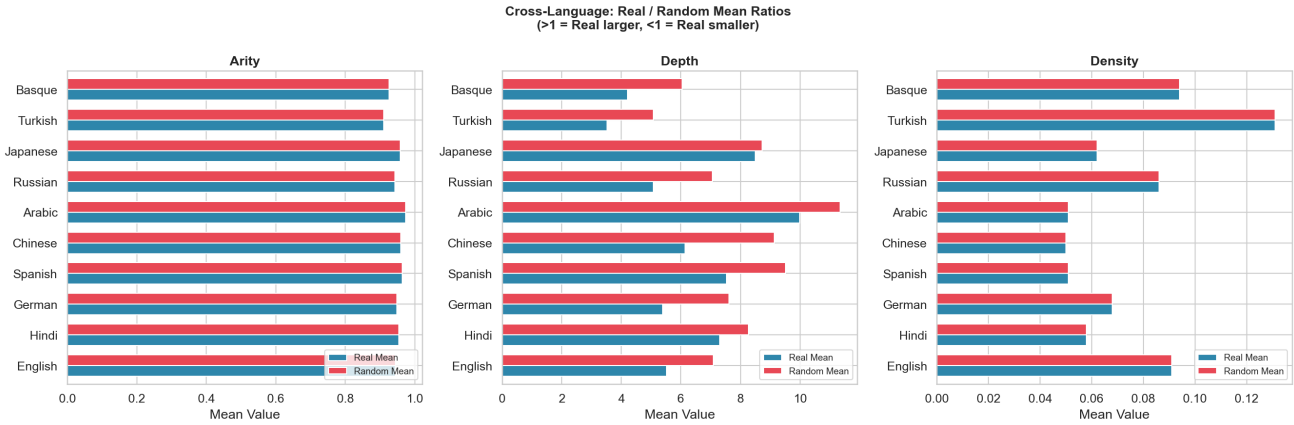

Figure 2: Cross-language mean arity, depth, and density.

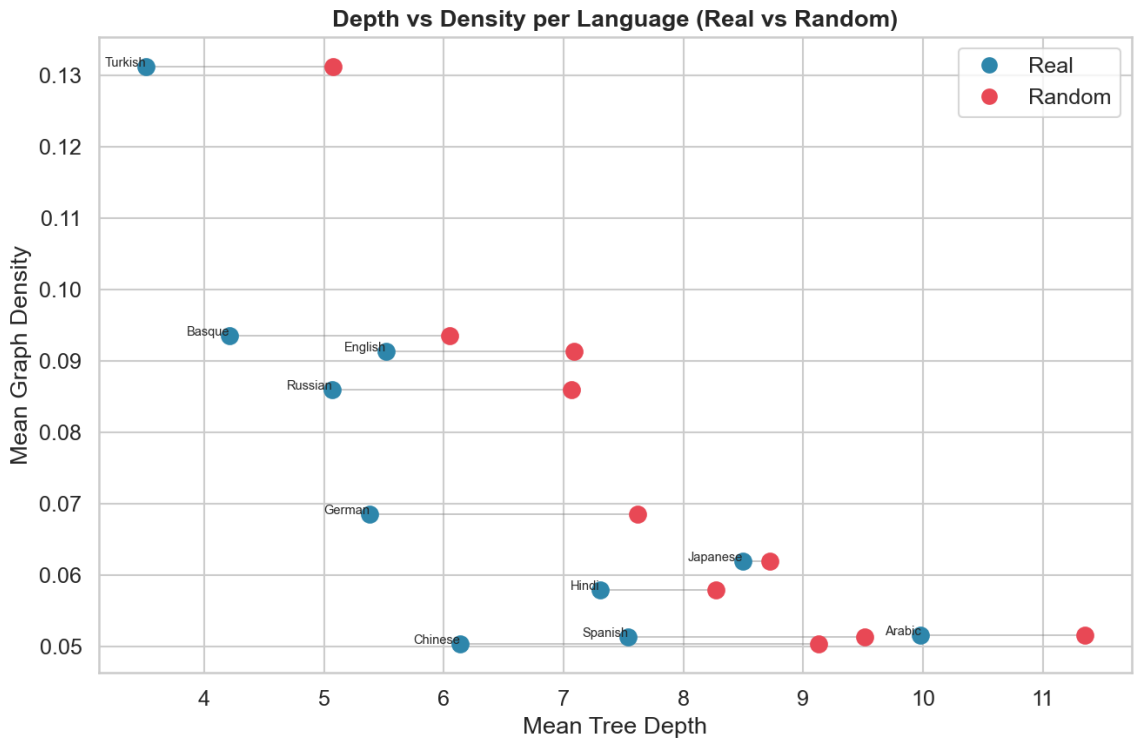

Figure 3: Depth vs. density scatter per language.

4.4 Length-Controlled Depth Analysis. A potential confound in the depth comparison is sentence length: if real and random trees differ in their sentence length distributions, the observed depth gap could be a length artifact rather than a genuine structural property. To rule this out, we binned all sentences into five length brackets (4–7, 8–11, 12–15, 16–20, and 21+ words) and compared mean real vs. random depth *within each bin separately*. Within each bin, sentence lengths are approximately equal, so any remaining depth difference must reflect structural organisation rather than size.

As shown in Figures 4 and 5, the depth gap persists robustly at *every* length bin in *every* language, with only two exceptions: Arabic (8–11 and 12–15 words) and Japanese (12–15 and 21+ word bins), both of which are consistent with the typological outlier status of these languages noted in Section 4.2. The gap grows monotonically with sentence length: for sentences of 4–7 words the mean gap is +0.87 depth units, rising to +2.44 for sentences of 21+ words. This indicates that longer sentences show *proportionally greater* depth suppression in real language, consistent with the working memory account cognitive pressure to avoid deep nesting intensifies as sentence complexity increases.

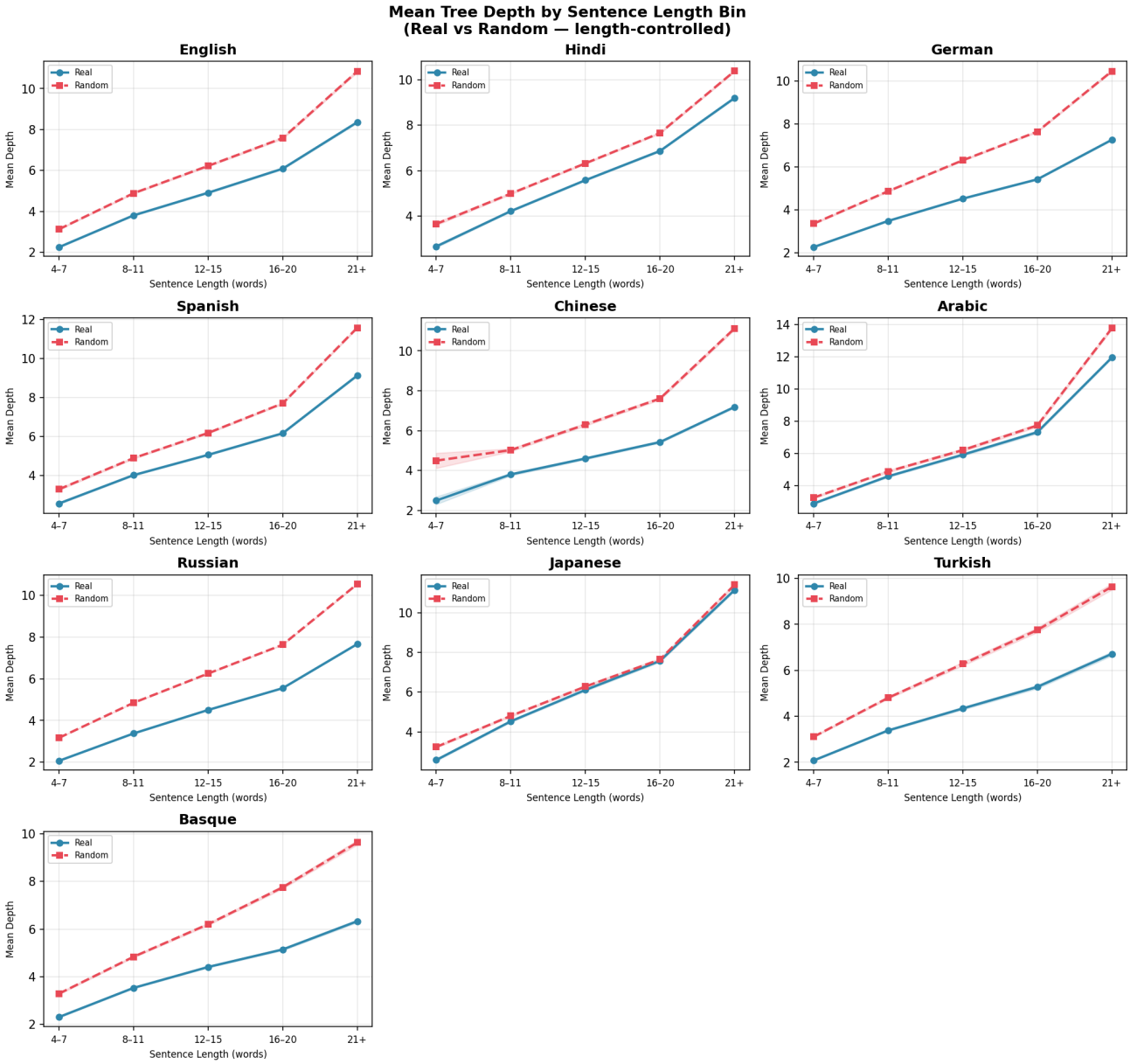

Figure 4: Mean tree depth per sentence length bin for each language (real = blue, random = red). Real trees are consistently shallower than random trees at every length bracket across all languages.

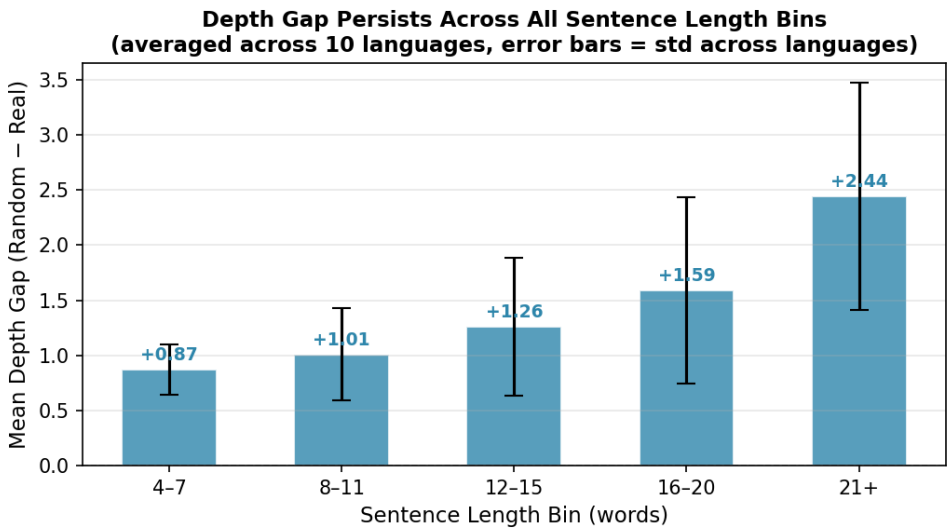

Figure 5: Mean depth gap (random − real) per length bin, averaged across all 10 languages. Error bars show standard deviation across languages. The gap is positive at every bin and increases monotonically with sentence length.

5. Theoretical Implications

The central finding that natural language dependency trees are consistently shallower than random trees of the same size has direct implications for both language processing and language evolution theories.

Cognitive constraints on depth. The working memory hypothesis of language processing predicts that deeply nested structures should be avoided because they require the parser to hold multiple incomplete dependencies in memory simultaneously. Our results support this directly: across ten typologically different languages, depth is suppressed relative to what random assignment would produce. This is consistent with Dependency Length Minimization (DLM) theory [1], which posits that languages tend to arrange words so as to keep syntactically related words close together, naturally producing flatter trees.

Japanese as a typological outlier. The near-zero depth gap for Japanese (3%) suggests that its head-final, verb-final typology permits or requires deeper nesting in a way other languages do not. This is not evidence against the cognitive constraint theory, but rather evidence that the constraint interacts with the morphosyntactic affordances of each language type. Morphologically rich, agglutinative languages like Japanese may offload structural complexity onto morphology rather than word order, permitting deeper syntactic chains.

Polarised arity as a signature of grammar. The consistent difference in arity distributions real trees being more polarised reflects the categorical nature of grammatical roles. In any language, most function words (determiners, auxiliaries, case markers) are leaves, while a small class of predicates governs many arguments. Random trees have no such categorical distinction, producing smoother branching distributions. This polarisation may be a universal signature of compositional grammar.

Implications for language evolution. These findings suggest that the structural properties of human language dependency trees are not accidental but reflect pressures that likely shaped the evolution of syntactic structure: the need to communicate compositionally while remaining within the limits of real-time comprehension. Any theory of language evolution must account for why natural syntax consistently produces structures shallower than the random baseline, across languages as typologically distant as Turkish, Arabic, Japanese, and English.

Implications for computational parsing. The structural properties identified here have direct consequences for natural language processing systems. Dependency parsers both transition-based (shift-reduce) and graph-based neural models have time and memory complexity that scales with tree depth and maximum arity. Shallower real trees mean that transition-based parsers encounter fewer long-range dependencies, reducing the number of steps the parser must hold an open dependency in its stack [12]. This may partly explain why greedy transition-based parsers achieve surprisingly high accuracy despite their limited lookahead: the structural properties of human language trees keep most decisions locally resolvable. The polarised arity distribution reinforces this: since most nodes are leaves (arity 0), the majority of parsing decisions involve attaching a terminal word rather than resolving a complex branching structure. Conversely, the Japanese outlier where real trees approach random depth at longer sentences is consistent with the well-documented difficulty of Japanese dependency parsing, particularly for long sentences with centre-embedding [14]. These findings suggest that linguistic universals of tree shallowness are not merely cognitively convenient but are also a key reason why statistical and neural parsers generalise well across languages: the hypothesis space of plausible trees is far smaller than the full space of random DAGs.

References

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Appendix A: Code

Full code is available on GitHub: https://github.com/AkarshVerma737/Empirical-Properties-of-Directed-Acyclic-Graphs-in-Natural-Language.git

depgraph.py — CoNLL-U parsing to NetworkX DiGraph

```
def tokens_to_digraph(tokens):
    G = nx.DiGraph()
    root = None
    for tok in tokens:
        G.add_node(tok.id, form=tok.form, upos=tok.upos)
    for tok in tokens:
        if tok.head == 0:
            root = tok.id
            G.graph["root"] = tok.id
        else:
            G.add_edge(tok.head, tok.id, deprel=tok.deprel)
    if root is None or not nx.is_weakly_connected(G):
        return None
    return G
```

treegen.py — Prüfer sequence random tree generation

```
def random_rooted_tree(n, root=0):
    if n <= 2:
        G = nx.DiGraph()
        if n == 2: G.add_edge(0, 1)
        else: G.add_node(0)
        G.graph["root"] = 0
        return G
    prufer_seq = np.random.randint(0, n, size=n-2).tolist()
    degree = [1] * n
    for node in prufer_seq:
        degree[node] += 1
    edges = []
    for node in prufer_seq:
        for leaf in range(n):
            if degree[leaf] == 1:
                edges.append((node, leaf))
                degree[node] -= 1
                degree[leaf] -= 1
                break
    last = [i for i in range(n) if degree[i] == 1]
    edges.append(tuple(last))
    T = nx.Graph()
    T.add_nodes_from(range(n))
    T.add_edges_from(edges)
    return nx.bfs_tree(T, source=root)
```

compute_metrics.py — Arity, Depth, Density

```
def compute_arity(G):
    return [d for _, d in G.out_degree()]

def compute_depth(G):
    root = G.graph.get("root")
    queue = [(root, 0)]
    visited = set()
    max_d = 0
    while queue:
        node, d = queue.pop(0)
        if node in visited: continue
        visited.add(node)
        max_d = max(max_d, d)
        for child in G.successors(node):
            queue.append((child, d + 1))
    return max_d

def compute_density(G):
    n = G.number_of_nodes()
    return G.number_of_edges() / (n * (n - 1)) if n > 1 else 0.0
```

analysis.py — KS Test and Summary Statistics

```
from scipy.stats import ks_2samp

def ks_test(real, random, metric_name):
    stat, p = ks_2samp(real, random)
    return {"metric": metric_name, "ks_stat": round(stat, 4),
            "p_value": round(p, 6), "significant": p < 0.05}

def run_analysis(real_agg, rand_agg, language):
    results = {"language": language, "ks_tests": []}
    for metric, key in [("arity", "all_arities"), ("depth", "depths"),
                        ("density", "densities")]:
        results["ks_tests"].append(ks_test(real_agg[key], rand_agg[key], metric))
    return results
```

Appendix B: Per-Language Distribution Plots

Violin plots of arity, tree depth, and graph density for each language, comparing real trees (blue) against size-matched random trees (red). The inner box-plot shows the median and IQR.

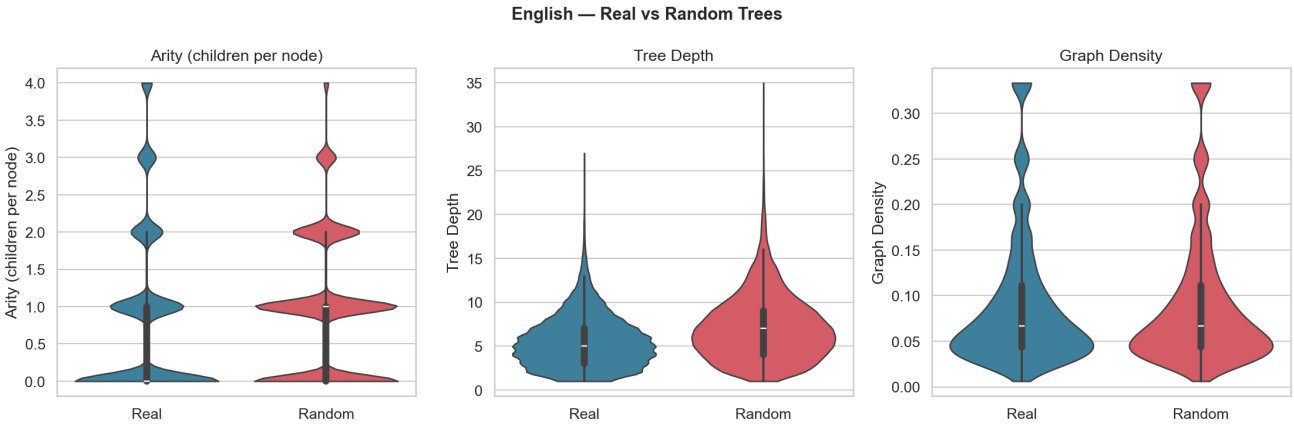

Figure 6: English: Depth reduction: 22%. KS stat (depth) = 0.184.

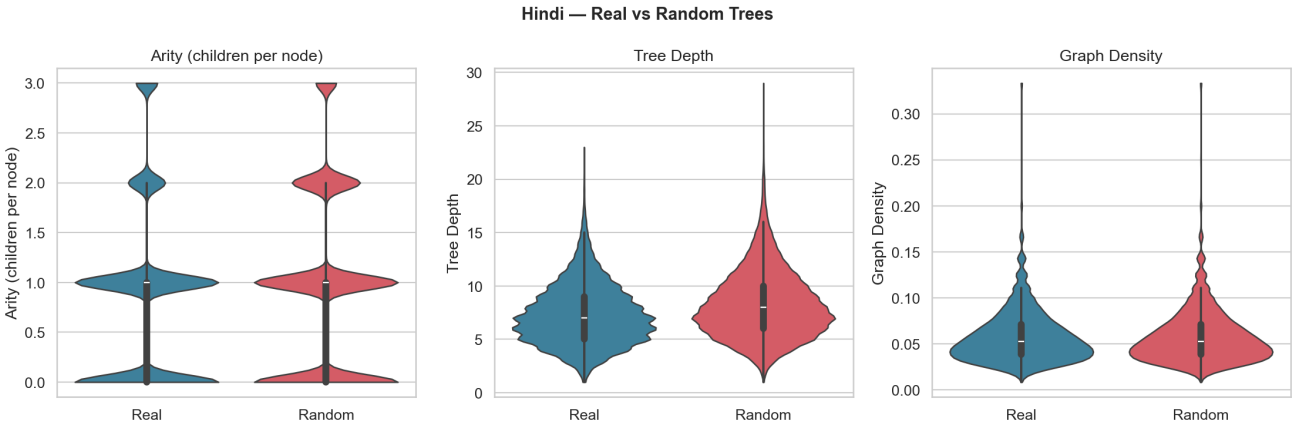

Figure 7: Hindi: Depth reduction: 12%. KS stat (depth) = 0.118.

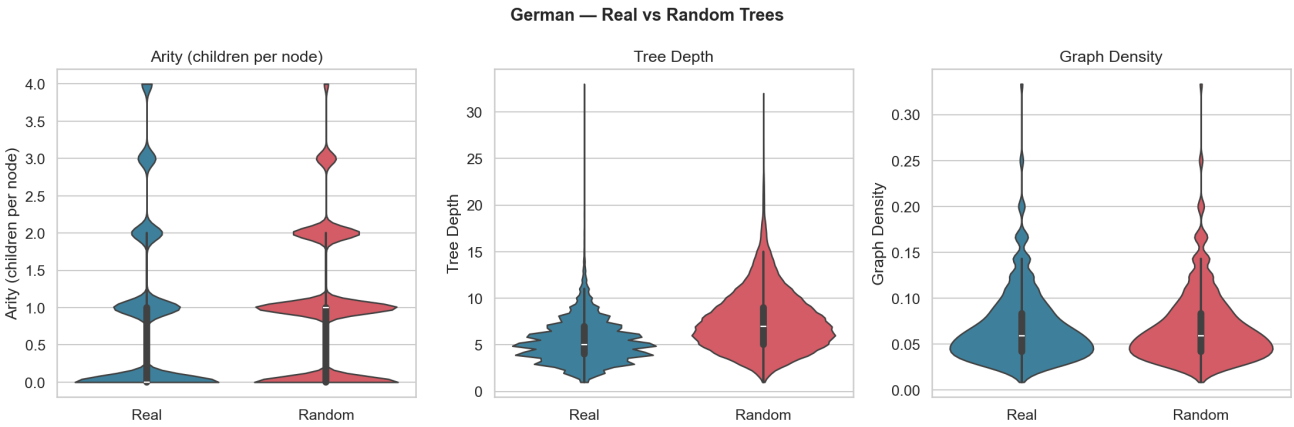

Figure 8: German: Depth reduction: 29%. KS stat (depth) = 0.320.

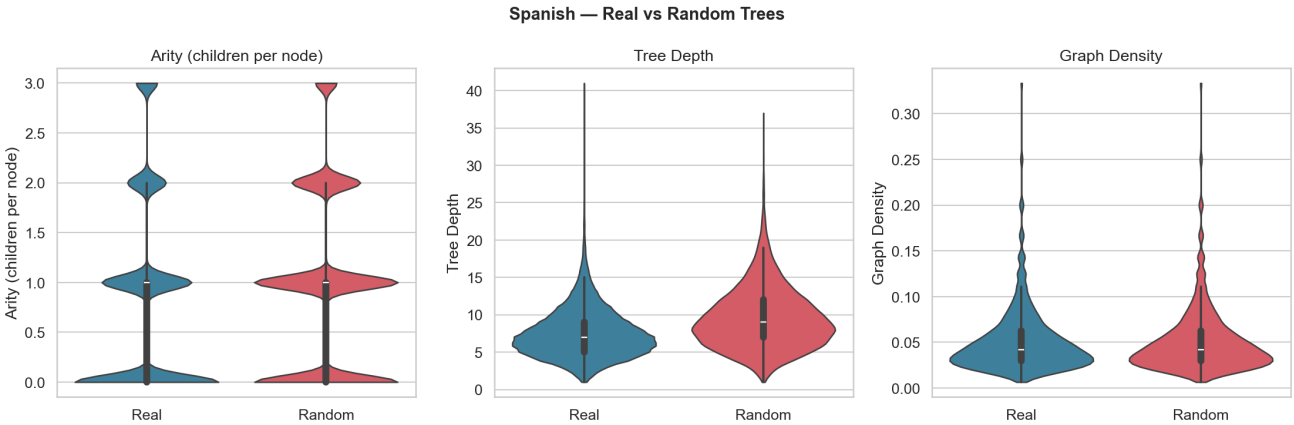

Figure 9: Spanish: Depth reduction: 21%. KS stat (depth) = 0.222.

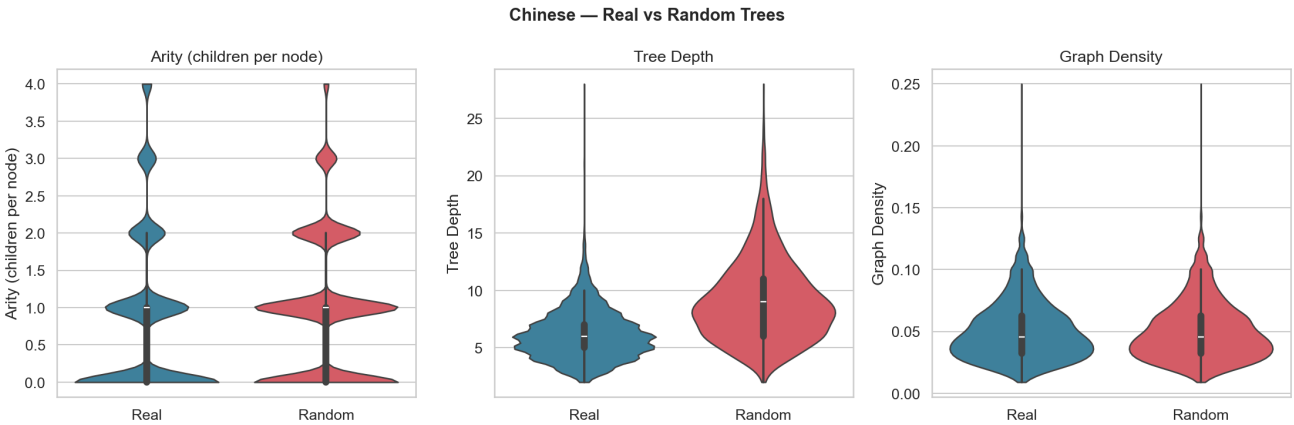

Figure 10: Chinese: Depth reduction: 33%. KS stat (depth) = 0.406.

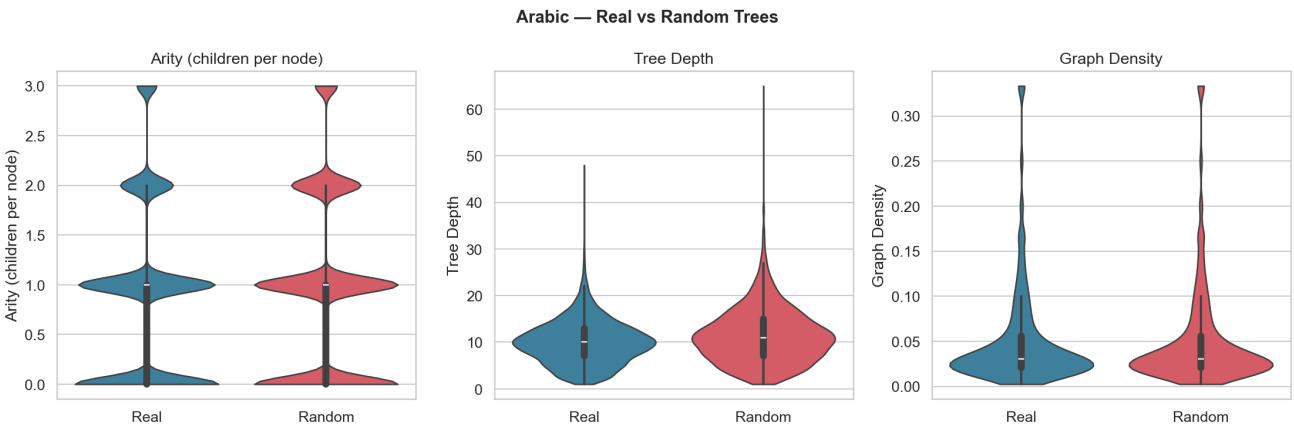

Figure 11: Arabic: Depth reduction: 12%. KS stat (depth) = 0.112.

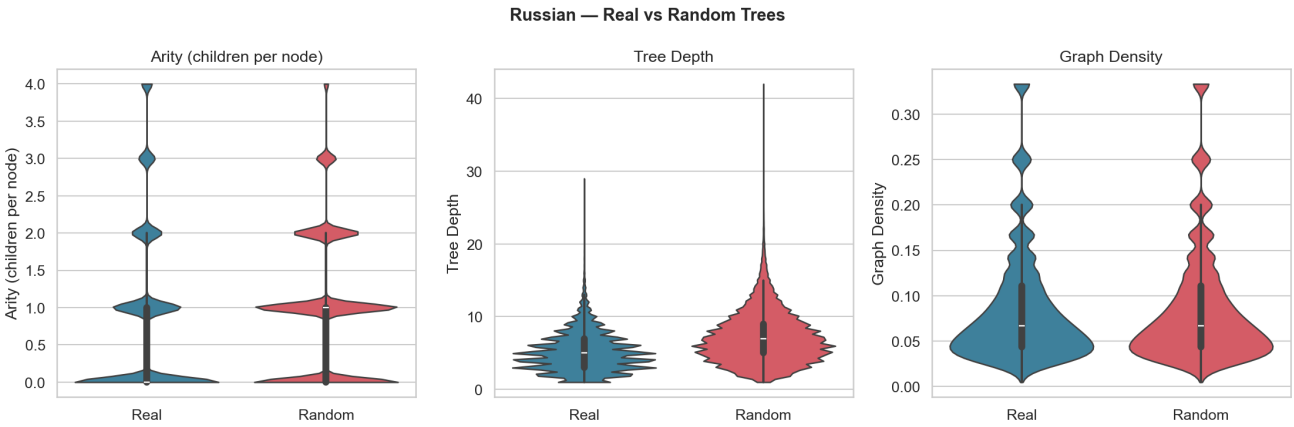

Figure 12: Russian: Depth reduction: 28%. KS stat (depth) = 0.247.

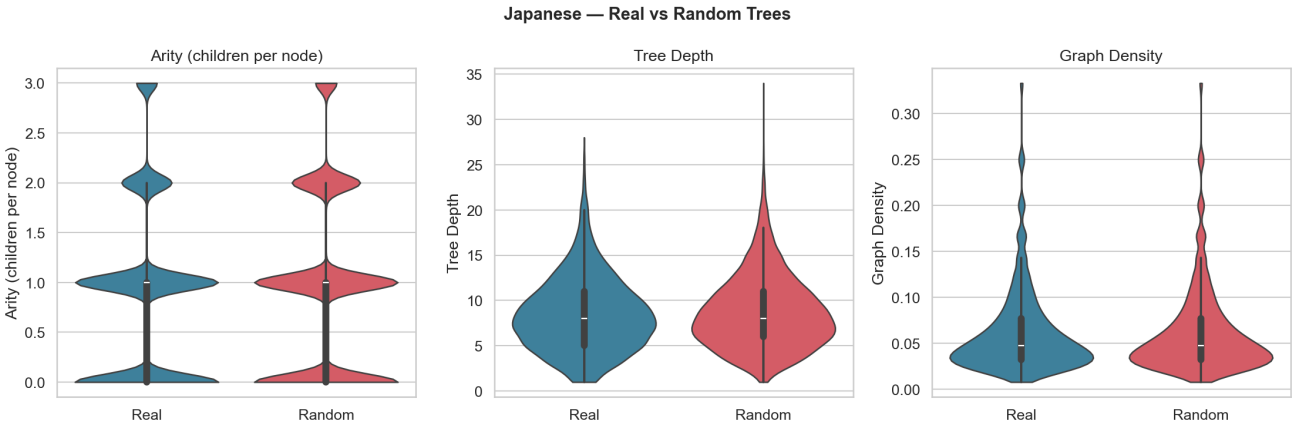

Figure 13: Japanese — Depth reduction: 3% (smallest gap across all languages). KS stat (depth) = 0.028, $p = 0.008$.

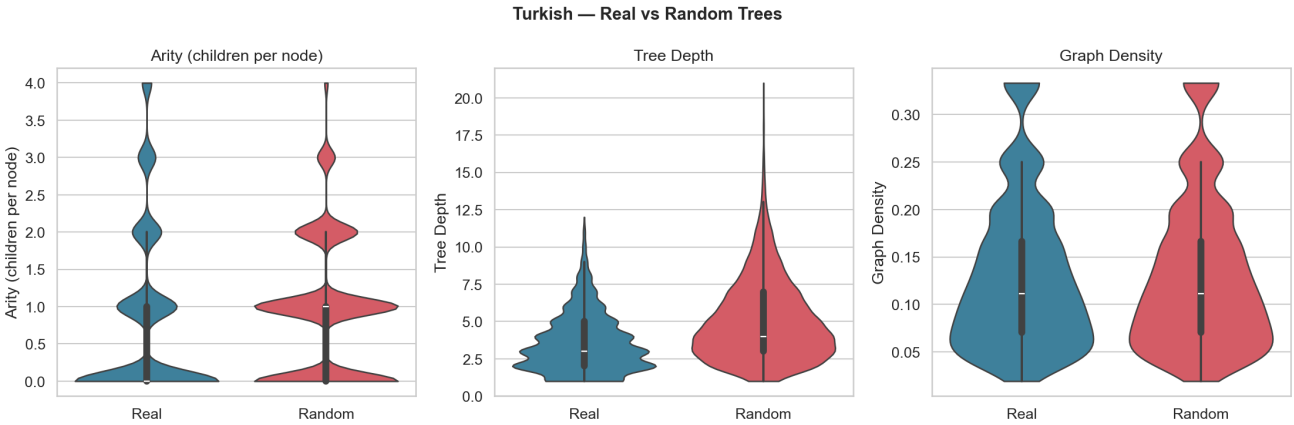

Figure 14: Turkish: Depth reduction: 31%. KS stat (depth) = 0.242.

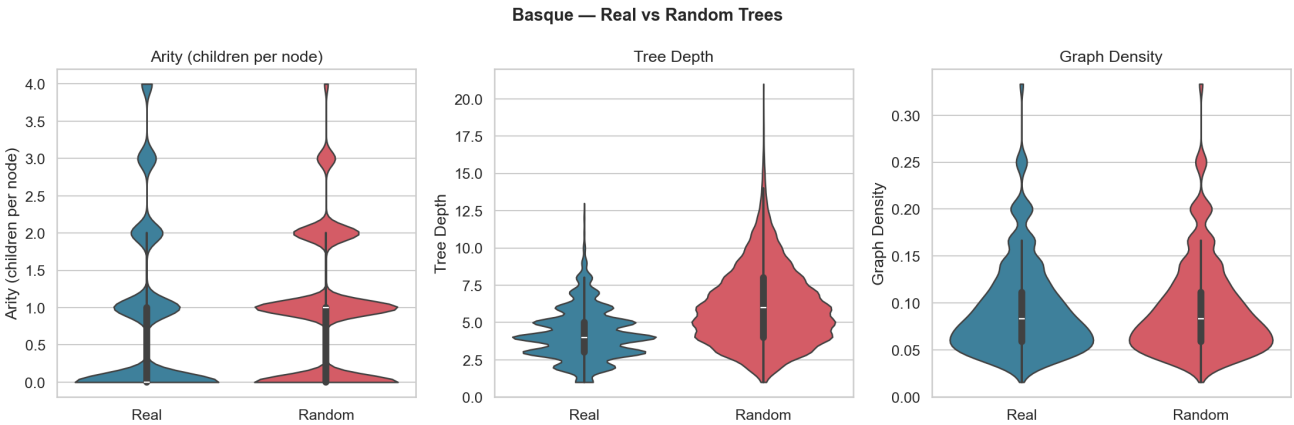

Figure 15: Basque: Depth reduction: 30%. KS stat (depth) = 0.330.